

A guide for antihypertensive medications for children on hemodialysis

1-Blood pressure should be checked for all children before hemodialysis. In children with blood pressure < 95th percentile, all anti-hypertensive medications should be held before hemodialysis. An individualized plan regarding antihypertensive medication should be documented in the notes for each patient on admission.

2-In children with blood pressure at or above 95th percentile, due short acting anti-hypertensive medications (like hydralazine, prazosin and propranolol) should be given regardless of time of hemodialysis. If a medication is due at time of hemodialysis, it should be administered after hemodialysis. Long-acting antihypertensive medications (like amlodipine and atenolol) should be held before hemodialysis.

3-For each patient on hemodialysis and on anti-hypertensive medications, BP percentiles should be checked and documented in the notes for each admission.

4- If an antihypertensive medication is held prior to hemodialysis, it should be given post hemodialysis when the blood pressure is above the limit provided by the treating team.

5-Some children have high BP during dialysis (intra-dialytic hypertension). In these children, antihypertensive medications are specifically ordered to be given before hemodialysis and this should be stated clearly by the treating team.

6-Administering anti-hypertensive medications prior to dialysis should not result in delay of haemodialysis unless the child is hypotensive or has other reasons to delay haemodialysis.

7- Children should be prioritized for hemodialysis. Elective planned hemodialysis sessions for children should be done during the morning time as soon as a slot for hemodialysis is available.

8-Short acting intravenous antihypertensive medication (hydralazine) should be kept available in the hemodialysis unit in case it is needed for a child who has high blood pressure during hemodialysis. Nurses in the hemodialysis unit should ensure the availability of intravenous hydralazine.

Some anti-hypertensive medications are cleared by hemodialysis and preferably given after hemodialysis.

Examples of anti-hypertensive medication removed by haemodialysis:

ACE inhibitors:

Captopril
Enalapril
Lisinopril
Ramipril
Cilazapril
Perindopril

Beta Blockers:

Atenolol
Metoprolol
Nadolol

Others:

Methyldopa
Minoxidil
Clonidine

Examples of anti-hypertensive medication NOT removed by haemodialysis:**ACE inhibitors:**

Fosinopril

Calcium channel blockers:

Amlodipine
Nifedipine

Beta blockers:

Labetalol
Carvedilol

Angiotensin receptor blockers:

Losartan
Valsartan

Others:

Hydralazine
Prazosin

Examples of short-acting antihypertensive medications:

Enalapril
Hydralazine
Prazosin
Propranolol

Examples of long- acting antihypertensive medications:

Amlodipine
Atenolol
Lisinopril